Predicting and Clustering National Development Trajectories from Health and Economic Indicators

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Abstract

We assemble an original country–year panel (217 countries, 2005–2020) by joining the World Bank's *World Development Indicators* with the *World Happiness Report* (Gallup World Poll), and use it to ask two questions. **(1) Prediction:** given a country's economic, health and social indicators in year *t*, how well can we forecast its life expectancy and its subjective *life-ladder* score in year *t + 4*, and do social/health factors add predictive power on top of income? **(2) Clustering:** treating each country's multi-year path as a trajectory, do data-driven "development archetypes" line up with World Bank income tiers, or do they diverge? Using grouped-by-country cross-validation we find that a full economic + health + social feature set roughly **halves** life-expectancy forecast error versus an economics-only model (R^2 0.72 → 0.92) and that **social support**, not GDP, is the single strongest predictor of future happiness. Clustering countries on the *shape* of their trajectories recovers a robust developed/developing split (bootstrap ARI ≈ 0.96) whose main axis of disagreement with income tiers is **momentum**: a large group of low- and middle-income countries is improving far faster than their income class implies, while several resource-rich upper-middle-income states under-perform.

1. Problem statement and hypotheses

The "beyond GDP" movement argues that national income is a poor summary of human progress. We operationalise that debate as concrete machine-learning tasks.

Task 1 — Forecasting (supervised regression). For a country observed in year *t*, predict a key well-being outcome in year *t + k* (k = 4):

- y^{LE}_{t+k} — life expectancy at birth (a hard demographic outcome), and
- y^{LL}_{t+k} — the Cantril *life-ladder* score (subjective well-being, 0–10).

> **Hypothesis H1.** Economic inputs *alone* (GDP per capita, urbanisation, inequality) are weak predictors of future outcomes; adding health-system and social factors (immunisation, sanitation, physicians, social support, freedom) adds substantial predictive power.

Task 2 — Clustering (unsupervised). Represent each country by the *level, slope and volatility* of its indicators over time and group countries into a few development archetypes.

> **Hypothesis H2.** Data-driven trajectory clusters will *not* simply reproduce the four World Bank income tiers; the mismatches identify countries that over- or under-perform their income class.

Why machine learning? A database query returns today's indicator values but cannot map a nonlinear combination of many correlated inputs to a *future* outcome, nor rank which present-day factors *precede* health gains — both require a fitted predictive model and its feature-attribution. Likewise there is no ground-truth "development type" to `GROUP BY`; the archetypes must be *discovered* in high-dimensional trajectory space. The value comes from generalisation and pattern discovery, not from summarising values that already exist in a table.

2. Data and methods

All figures and tables below are produced by the pipeline in `src/` and regenerated by `python run_all.py`. This notebook loads the cached artifacts from `figures/` and `results/` so the report renders quickly and deterministically.

```
Panel: 3472 country-year rows, 217 countries, years 2005-2020
```

2.1 Data sources and integration

We build the panel ourselves from three public sources joined on ISO-3 country code and year:

Source	Access	Variables
World Bank WDI	`wbgapi` client	life expectancy, under-5 & infant mortality, fertility, health expenditure p.c., DTP3 & measles immunisation, physicians/1k, sanitation & water access, GDP p.c. (PPP), urbanisation, primary enrolment, Gini
World Happiness Report (Gallup)	official "Data for Table 2.1" panel	life-ladder, log GDP p.c., social support, freedom, generosity, corruption perception, positive/negative affect
World Bank income tiers	`wbgapi` economy metadata	Low / Lower-middle / Upper-middle / High income, region

WHR country names are reconciled to World Bank ISO-3 codes with a curated name-fix map (e.g. *Russia → Russian Federation*, *Czech Republic → Czechia*); only three WHR entities that are not World Bank economies (North Cyprus, Somaliland region, Taiwan) are dropped. The merge yields **3,472 country-year rows**.

Realism / sparsity. As anticipated, coverage is uneven: the Gini index and the subjective WHR variables are the sparsest, and the WHR only covers ~160 of the 217 countries. Pooling across country-years still yields a few thousand training rows, so we lean on regularisation and grouped cross-validation rather than high-variance models.

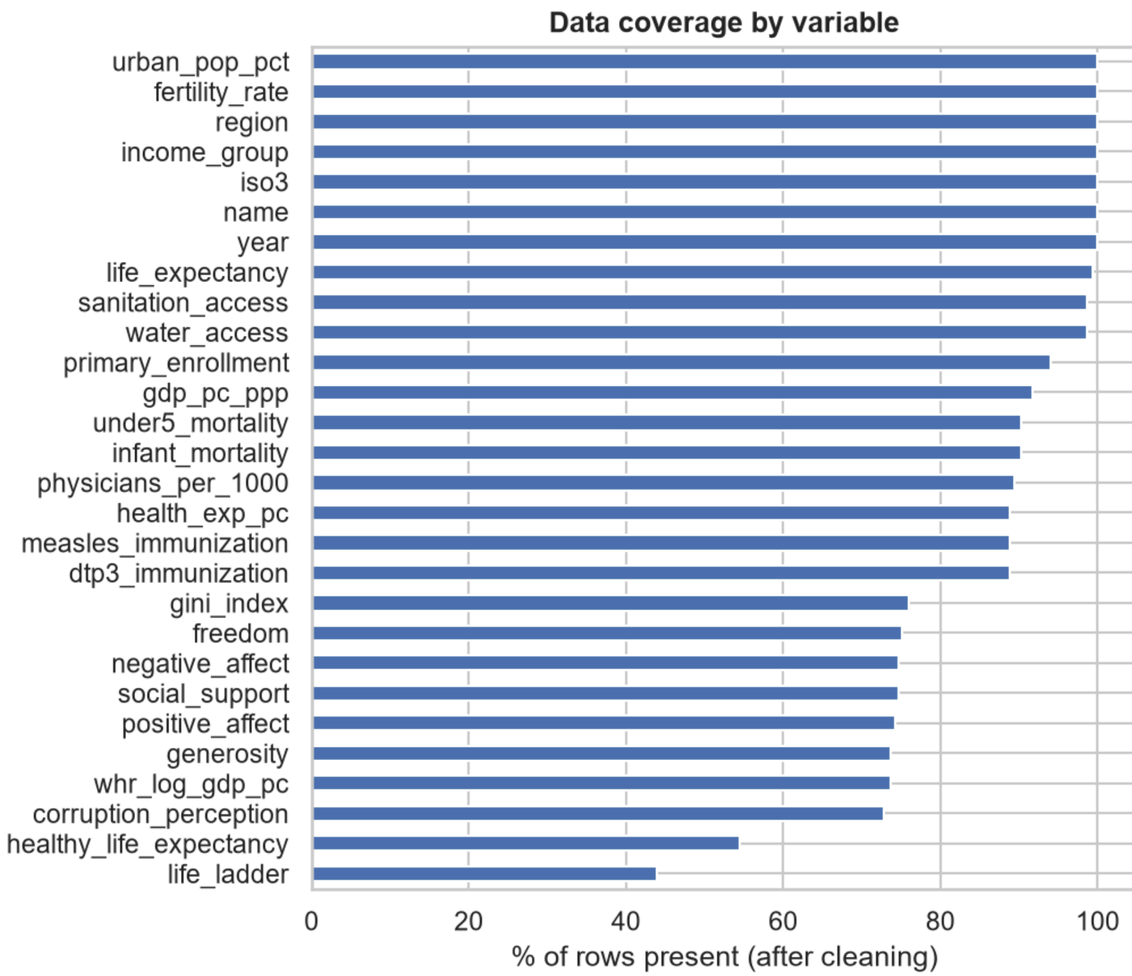
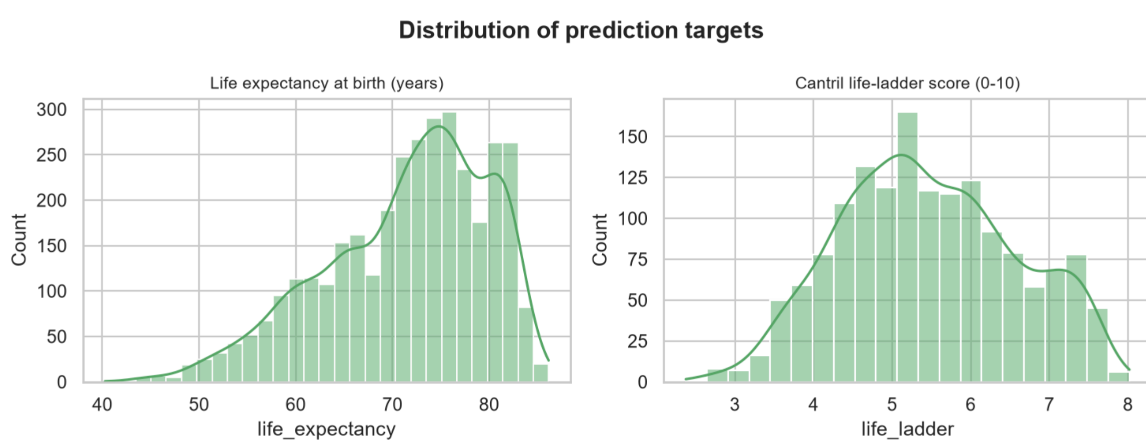


Figure 1. Variable coverage after cleaning. Hard demographic/economic indicators are near-complete; the Gini index and Gallup-sourced social variables are the sparsest, which motivates the imputation strategy below.



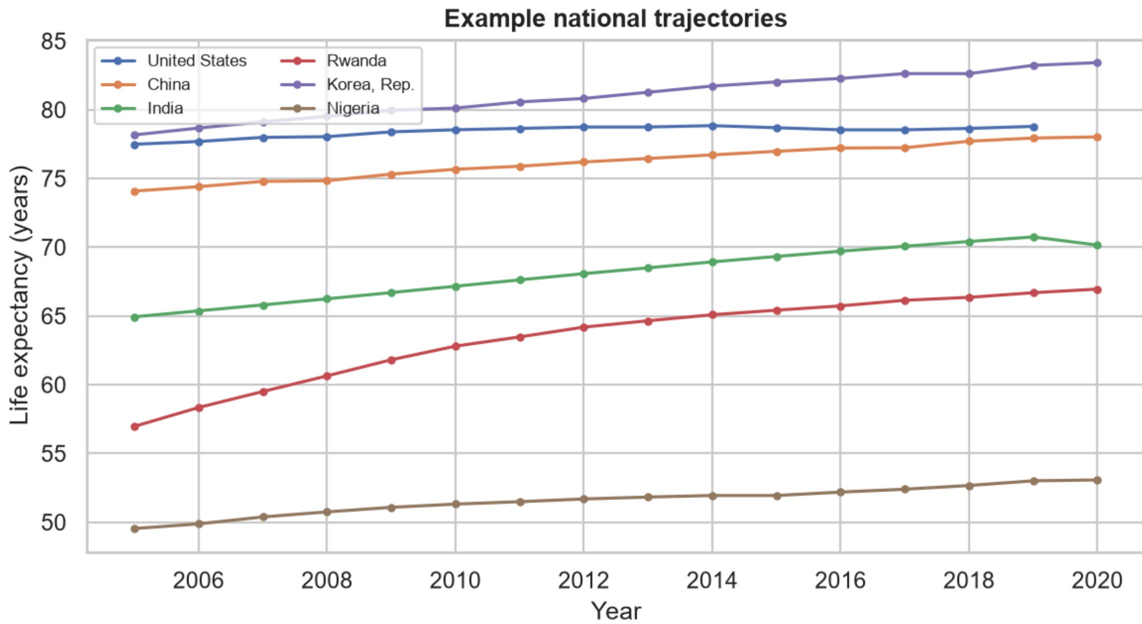


Figure 2. Prediction targets and example trajectories. Life expectancy is left-skewed (a long tail of low-life-expectancy country-years); the life-ladder is roughly symmetric. The trajectory panel shows why a *forecasting* framing is interesting: countries such as Rwanda and China rise steeply, others plateau.

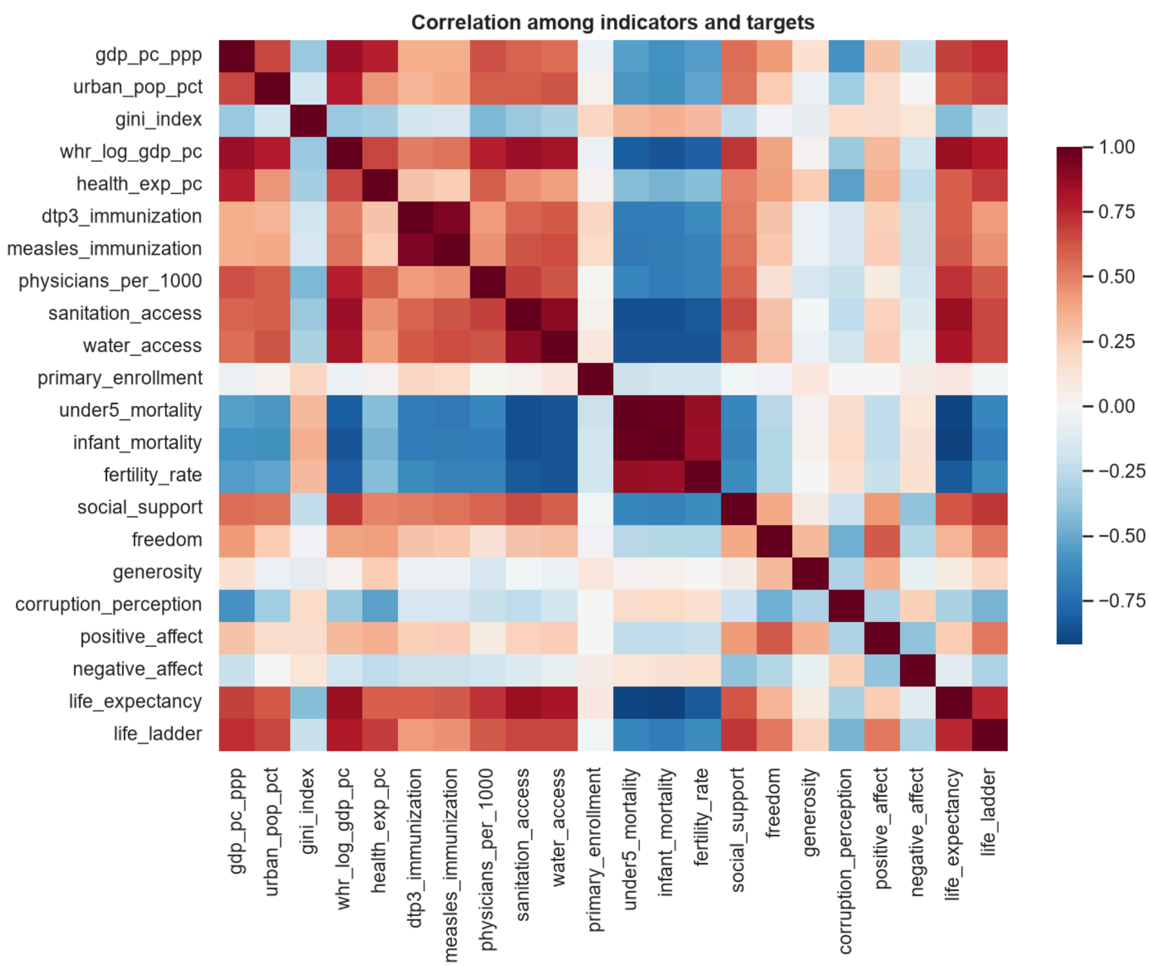


Figure 3. Correlation structure. Health and economic indicators are strongly inter-correlated (e.g. mortality ↔ life expectancy ↔ GDP), confirming that many inputs carry overlapping signal — a setting where regularised and tree-based models, plus permutation/SHAP attribution, are more trustworthy than raw coefficients.

2.2 Methodology

Cleaning & imputation. We despike obvious source errors with a robust within-country median/MAD rule (e.g. a Central African Republic life-expectancy value of 14.7 sandwiched between values near 50), interpolate interior gaps within each country's own series, and impute residual gaps with multivariate *IterativeImputer* (MICE) **inside** the model pipeline so it is fit only on training folds. Targets are never imputed — a row with a missing outcome is dropped, so no fabricated label enters training or evaluation.

Target construction. For each (country, *t*) we attach the outcome at *t + 4*. We model both the future **level** y_{t+k} and the multi-year **gain** $\Delta y = y_{t+k} - y_t$; the gain is the sharper test of *H1* because it asks which present conditions precede *improvement*.

Prediction models. Baseline linear regression, regularised Ridge and Lasso, and tree ensembles (Random Forest, XGBoost), each wrapped as `IterativeImputer → StandardScaler → estimator`. Every model's hyperparameters are tuned by **`GridSearchCV` under the same GroupKFold** (regularisation strength for Ridge/Lasso; depth, leaf size and number of trees for the Random Forest; depth, learning rate and number of rounds for XGBoost), scoring on RMSE; the selected settings are reported below.

Evaluation — leakage control. Our headline metric is **5-fold GroupKFold cross-validation grouped by country**: a country never appears in both train and test, so within-country autocorrelation cannot inflate scores. We also report a naive **persistence** baseline ($\hat{y}_{t+k} = y_t$) and a forward-in-time **temporal hold-out** (train on target years ≤ 2016, test 2017–2020).

Attribution. Permutation importance (on held-out years) and SHAP values rank which indicators drive the forecasts.

Clustering. Per country we engineer *level, slope (per-decade trend) and volatility* for eight broadly-covered indicators, standardise, and cluster with K-means and Ward hierarchical clustering. We choose *k* by silhouette + elbow, check stability by bootstrapping (Adjusted Rand Index), profile each cluster, and cross-tabulate against income tiers.

3. Results

3.1 Predicting the future from a country's conditions

Purpose. Establish how well each model forecasts the *t + 4* outcome from year-*t* indicators, judged against the persistence baseline under grouped-by-country CV.

```
=== Life expectancy (years)  —  5-fold GroupKFold (by country) ===
```

Model	RMSE	MAE	R2
Persistence (y_t)	1.585	1.191	0.962
Linear	2.990	2.230	0.864
Ridge	2.941	2.194	0.868
Lasso	2.990	2.228	0.864
Random Forest	2.764	2.017	0.884
XGBoost	2.651	1.928	0.893

```
=== Life-ladder score — 5-fold GroupKFold (by country) ===
```

Model	RMSE	MAE	R2
Persistence (y_t)	0.525	0.396	0.775
Linear	0.629	0.493	0.682
Ridge	0.616	0.484	0.695
Lasso	0.626	0.494	0.685
Random Forest	0.595	0.459	0.715
XGBoost	0.589	0.456	0.721

Hyperparameters selected by grouped-CV grid search (with the resulting cross-validated RMSE at those settings):

```
=== Life expectancy: tuned hyperparameters ===
```

Model	CV_RMSE	best_params
Linear	2.967	{}
Ridge	2.919	{'alpha': 100.0}
Lasso	2.965	{'alpha': 0.01}
Random Forest	2.738	{'max_depth': None, 'min_samples_leaf': 3, 'n_estimators': 400}
XGBoost	2.621	{'learning_rate': 0.03, 'max_depth': 4, 'n_estimators': 300, 'subsample': 0.8}

```
=== Life-ladder: tuned hyperparameters ===
```

Model	CV_RMSE	best_params
Linear	0.622	{}
Ridge	0.610	{'alpha': 100.0}
Lasso	0.617	{'alpha': 0.05}
Random Forest	0.590	{'max_depth': None, 'min_samples_leaf': 5, 'n_estimators': 400}
XGBoost	0.582	{'learning_rate': 0.03, 'max_depth': 6, 'n_estimators': 300, 'subsample': 0.8}

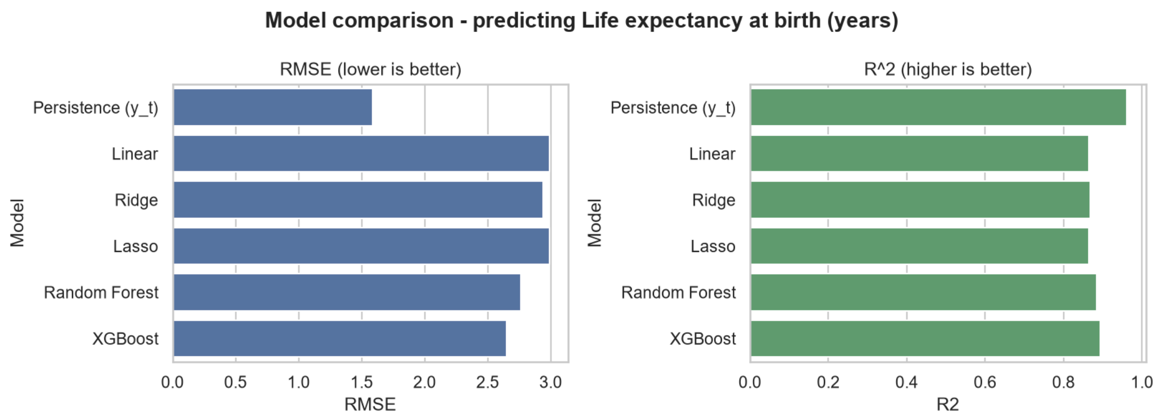


Figure 4 & Table 1. Model comparison. Two lessons stand out. First, **life expectancy is highly persistent**: the do-nothing baseline (R^2 = 0.96) is hard to beat over four years, and models that predict *from other conditions* *only* (no lagged outcome) trail it — XGBoost reaches R^2 = 0.89. This is the honest and important point: the interesting question is not "can we beat carry-forward" but "**which conditions carry information about the future**". Second, among learned models the **tree ensembles clearly dominate the linear family** even after tuning (XGBoost RMSE ≈ 2.65 vs. Ridge ≈ 2.94), evidence of genuine nonlinearity in the input→outcome map. When the current outcome *is* added as a feature, XGBoost rises to **R^2 = 0.981** (life expectancy) and **0.798** (life-ladder), exceeding persistence — so the indicators do add signal beyond simple carry-forward.

```
Autoregressive model (best model + current outcome), grouped CV:
  life_expectancy   R2=0.981  RMSE=1.109
  life_ladder       R2=0.794  RMSE=0.502
```

3.2 The central hypothesis (H1): does "beyond-GDP" information help?

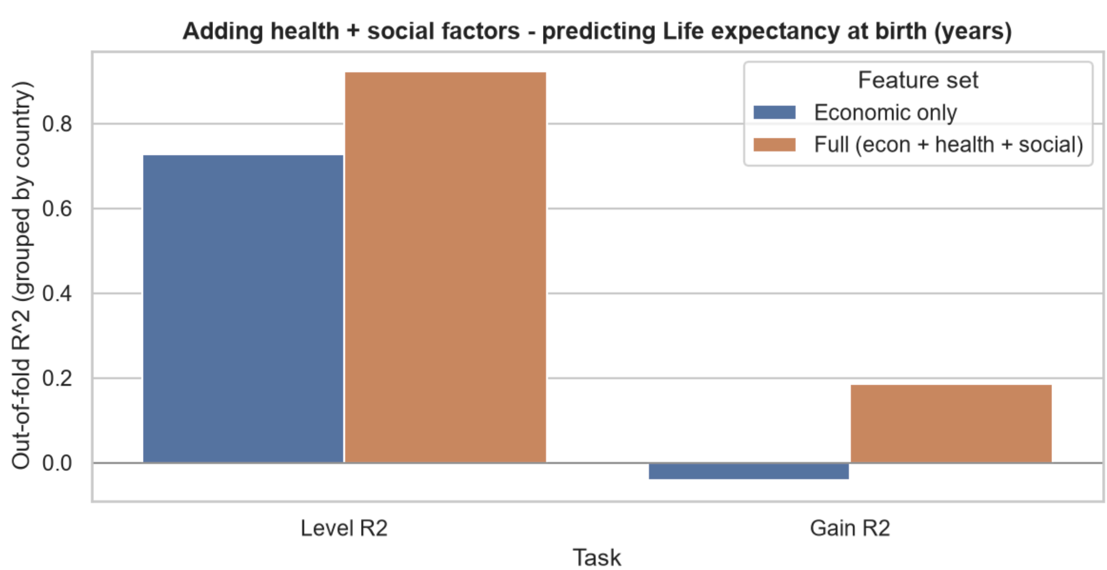
Purpose. On the *same* countries (those with observed social variables), compare an **economics-only** feature set against the **full** economic + health + social set, for both the future *level* and the multi-year *gain*.

```
=== Life expectancy: economics-only vs full feature set (grouped CV) ===
```

Feature set	n_features	Level RMSE	Level R2	Gain RMSE	Gain R2
Economic only	4	4.429	0.726	1.323	-0.042
Full (econ + health + social)	20	2.356	0.923	1.170	0.185

```
=== Life-ladder: economics-only vs full feature set (grouped CV) ===
```

Feature set	n_features	Level RMSE	Level R2	Gain RMSE	Gain R2
Economic only	4	0.722	0.581	0.569	-0.181
Full (econ + health + social)	18	0.589	0.721	0.535	-0.043



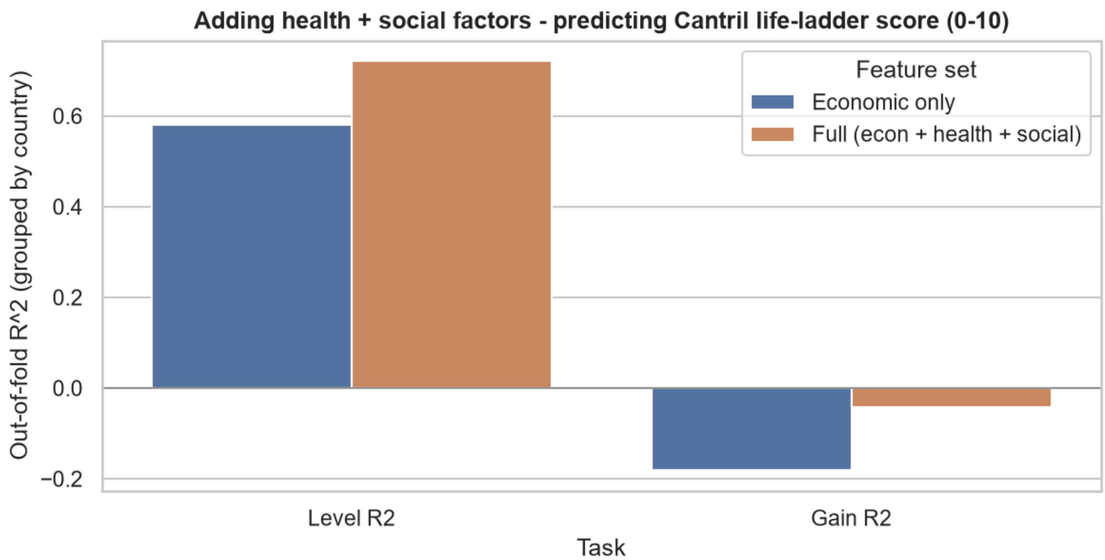
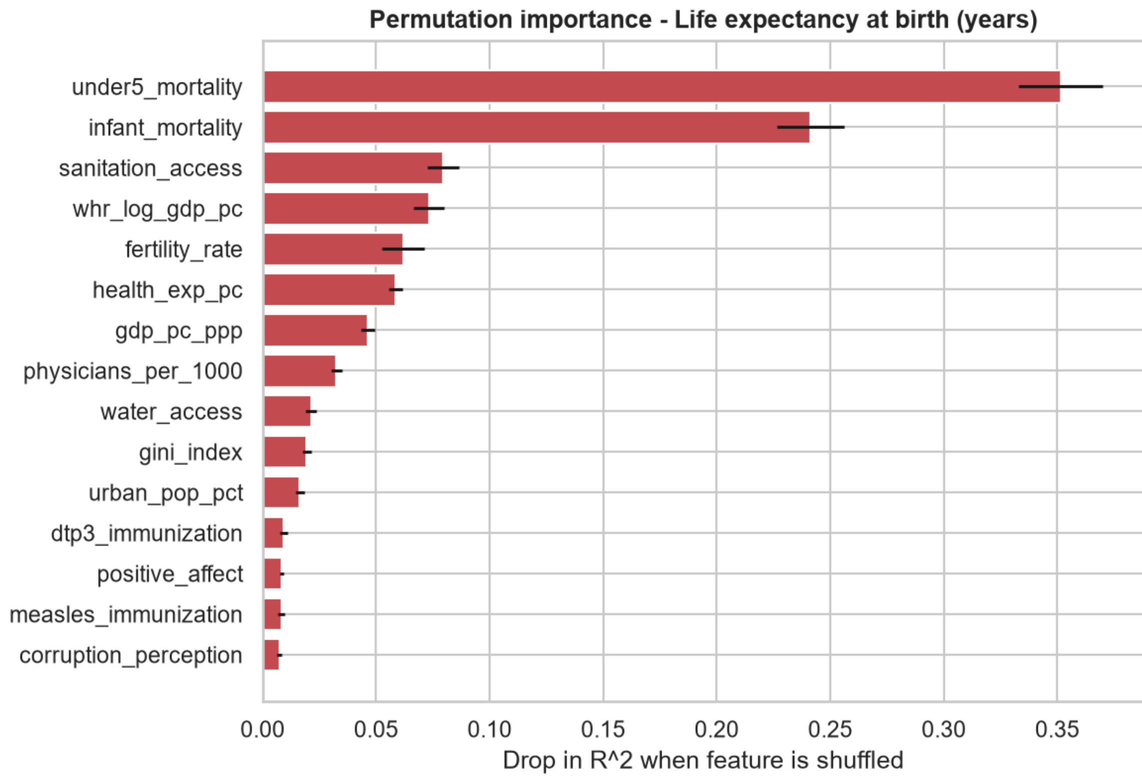


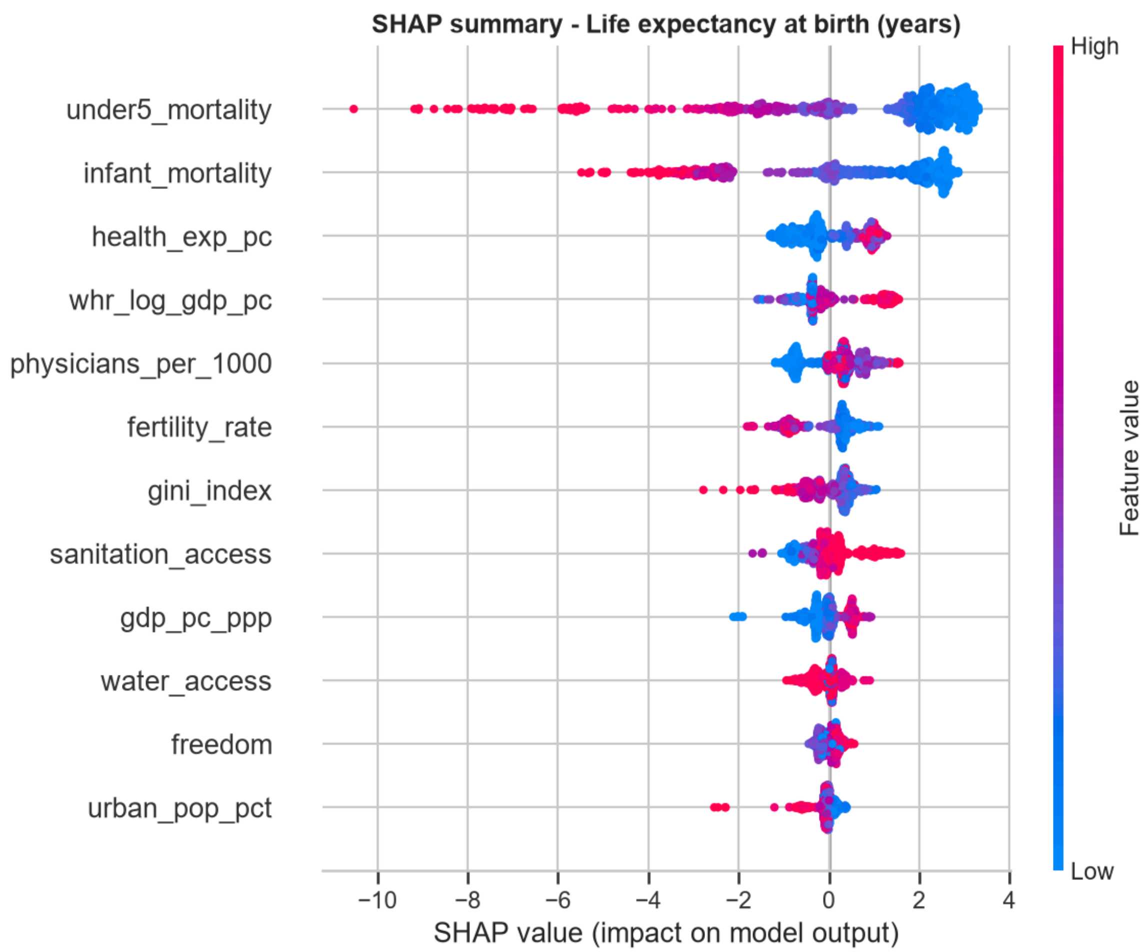
Figure 5 & Table 2. Hypothesis test. **H1 is strongly supported.** For life expectancy, moving from economics-only to the full set lifts the *level* R^2 from **0.72** → **0.92** and cuts RMSE almost in half (4.51 → 2.35 years). The effect is even sharper on the *gain* task, where economics-only is **worse than predicting the mean** ($R^2 = -0.10$) while the full set becomes informative ($R^2 = 0.17$). For the life-ladder, the full set raises level R^2 from **0.60** → **0.72**. Four-year *changes* in happiness are largely unpredictable from any of these slow-moving structural indicators (gain R^2 stays ≤ 0), a genuinely useful negative result: subjective well-being shifts are driven by shorter-term shocks our annual panel does not capture.

3.3 Which factors precede the outcomes?

Purpose. Rank indicators by permutation importance (drop in held-out R^2 when shuffled) and inspect SHAP value distributions for the XGBoost model.

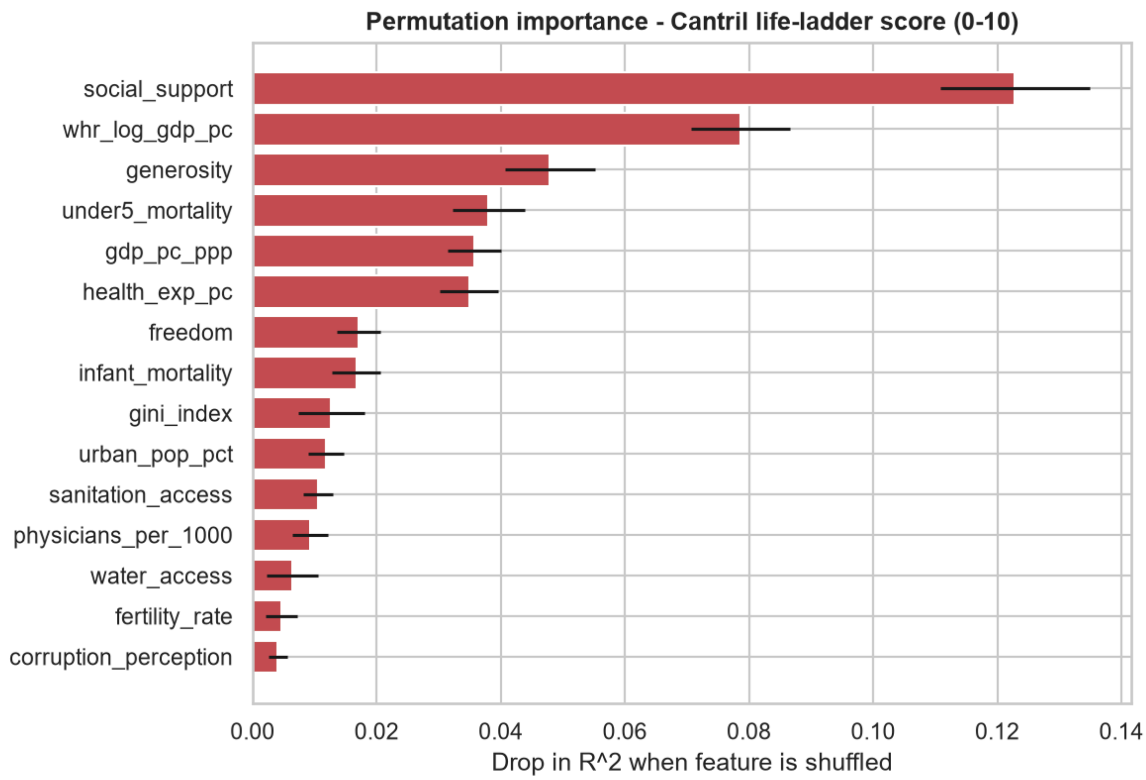


Permutation importance - Life expectancy at birth (years)
under5_mortality
infant_mortality
sanitation_access
whr_log_gdp_pc
fertility_rate
health_exp_pc
gdp_pc_ppp
physicians_per_1000
water_access
gini_index
urban_pop_pct
dtp3_immunization
positive_affect
measles_immunization
corruption_perception
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Drop in R^2 when feature is shuffled



SHAP summary - Life expectancy at birth (years)
under5_mortality
infant_mortality
health_exp_pc
whr_log_gdp_pc
physicians_per_1000
fertility_rate
gini_index
sanitation_access
gdp_pc_ppp
water_access
freedom
urban_pop_pct
−10
−8
−6
−4
−2
0
2
4
SHAP value (impact on model output)
High
Feature value
Low

Figure 6. Drivers of future life expectancy. Child-health outcomes dominate — **under-5 mortality** ($\Delta R^2 \approx 0.37$) and **infant mortality** (≈ 0.23) — followed by **sanitation access**, **fertility** and health spending. GDP per capita ranks mid-pack. The message aligns with the Preston-curve literature: once basic health conditions are accounted for, income adds relatively little to the *forecast* of future longevity.



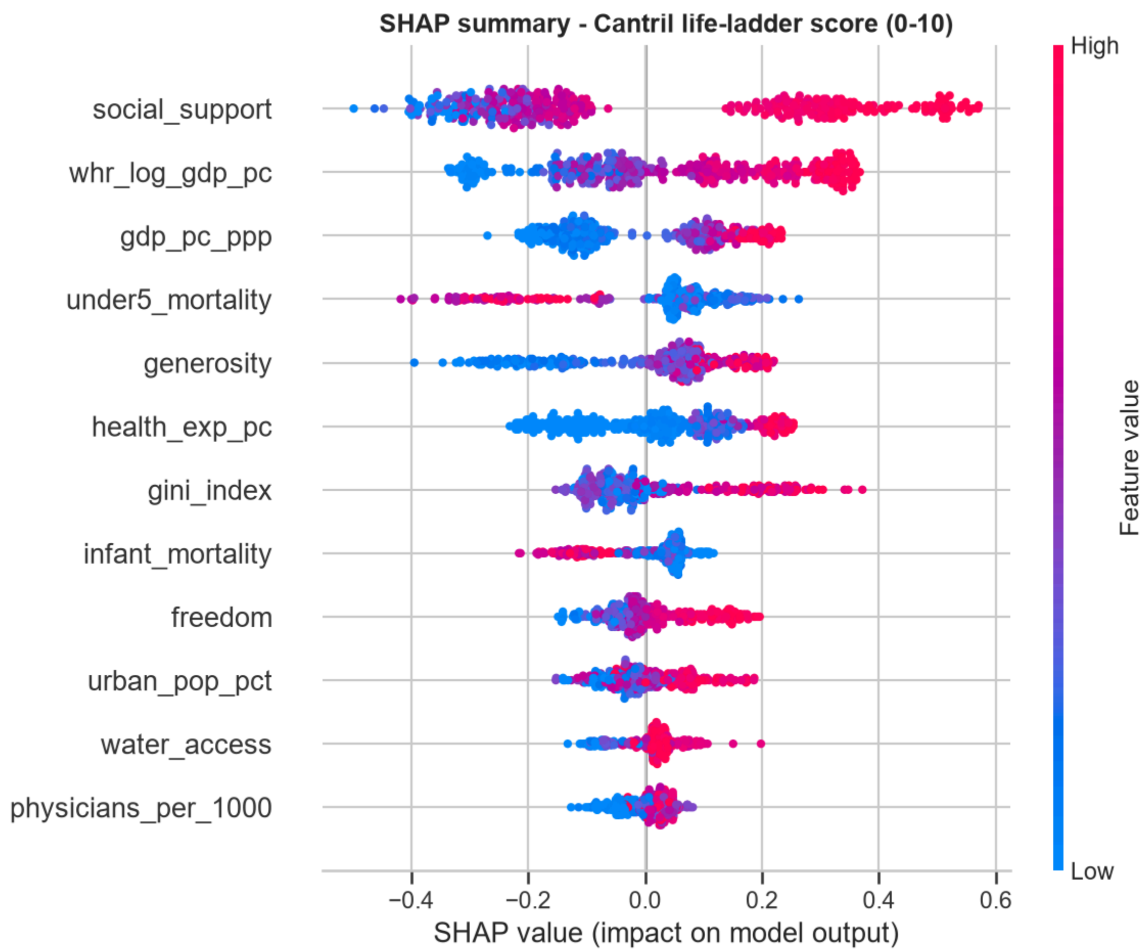


Figure 7. Drivers of future happiness. **Social support is the single strongest predictor** of the future life-ladder ($\Delta R^2 \approx 0.13$) — larger than (log) GDP per capita — with generosity and freedom also contributing. This is direct, model-based evidence for the "beyond GDP" thesis: for subjective well-being, the quality of a country's social fabric out-predicts its income.

3.4 Temporal generalisation

Purpose. Check forward-in-time generalisation: train on target years ≤ 2016, test on 2017–2020 (the same countries appear, unseen years).

```
=== Life expectancy: temporal hold-out (train ≤2016, test 2017–2020) ===
```

Model	RMSE	MAE	R2
Persistence (y_t)	1.367	0.976	0.968
Linear	2.780	1.996	0.867
Ridge	2.732	2.000	0.871
Lasso	2.783	2.001	0.866
Random Forest	1.669	1.196	0.952
XGBoost	1.780	1.267	0.945

```
=== Life-ladder: temporal hold-out (train ≤2016, test 2017–2020) ===
```

Model	RMSE	MAE	R2
Persistence (y_t)	0.578	0.437	0.709
Linear	0.602	0.464	0.696
Ridge	0.591	0.457	0.707
Lasso	0.596	0.469	0.702
Random Forest	0.543	0.422	0.753
XGBoost	0.518	0.396	0.775

Table 3. Temporal hold-out. Scores are **much higher** here than under grouped CV (XGBoost life-expectancy $R^2 = 0.95$ vs. 0.89). The gap is itself the insight: when a country has been *seen* (in earlier years) the model can exploit its level, so temporal generalisation is easy; generalising to *unseen countries* is the hard problem, and reporting both prevents the optimistic temporal number from overstating what the model has learned. Linear models fail to close the gap ($R^2 \approx 0.87$), again pointing to nonlinearity.

3.5 Clustering development trajectories (H2)

Purpose. Discover development archetypes from trajectory features and test whether they reproduce income tiers.

```
k-selection (silhouette + elbow):
```

	k	inertia	silhouette
0	2	3934	0.312
1	3	3565	0.300
2	4	3295	0.130
3	5	3137	0.107
4	6	2988	0.114
5	7	2853	0.109
6	8	2698	0.122

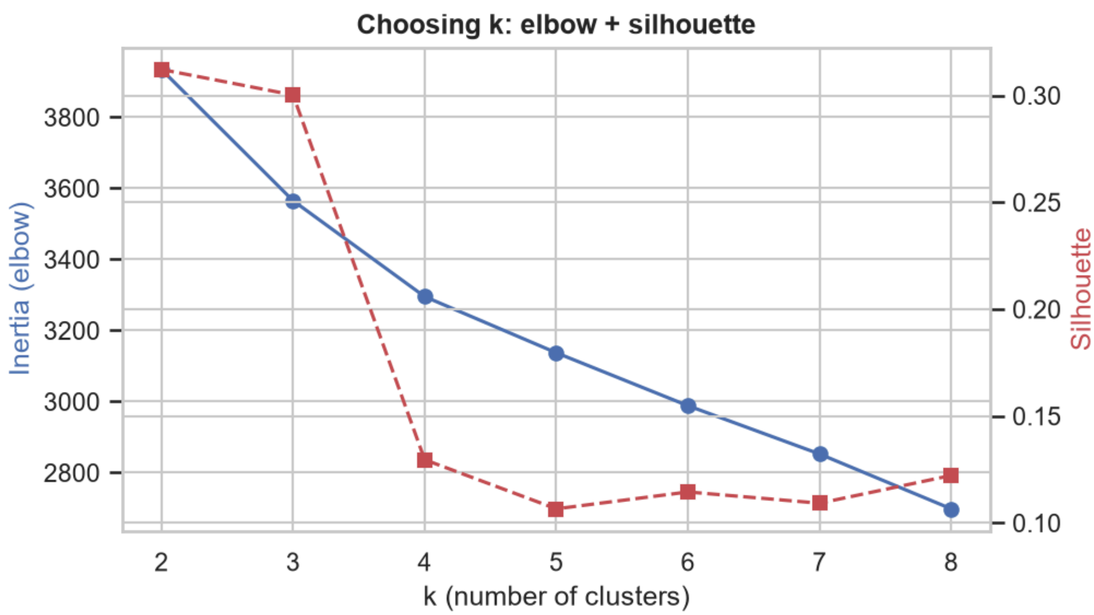
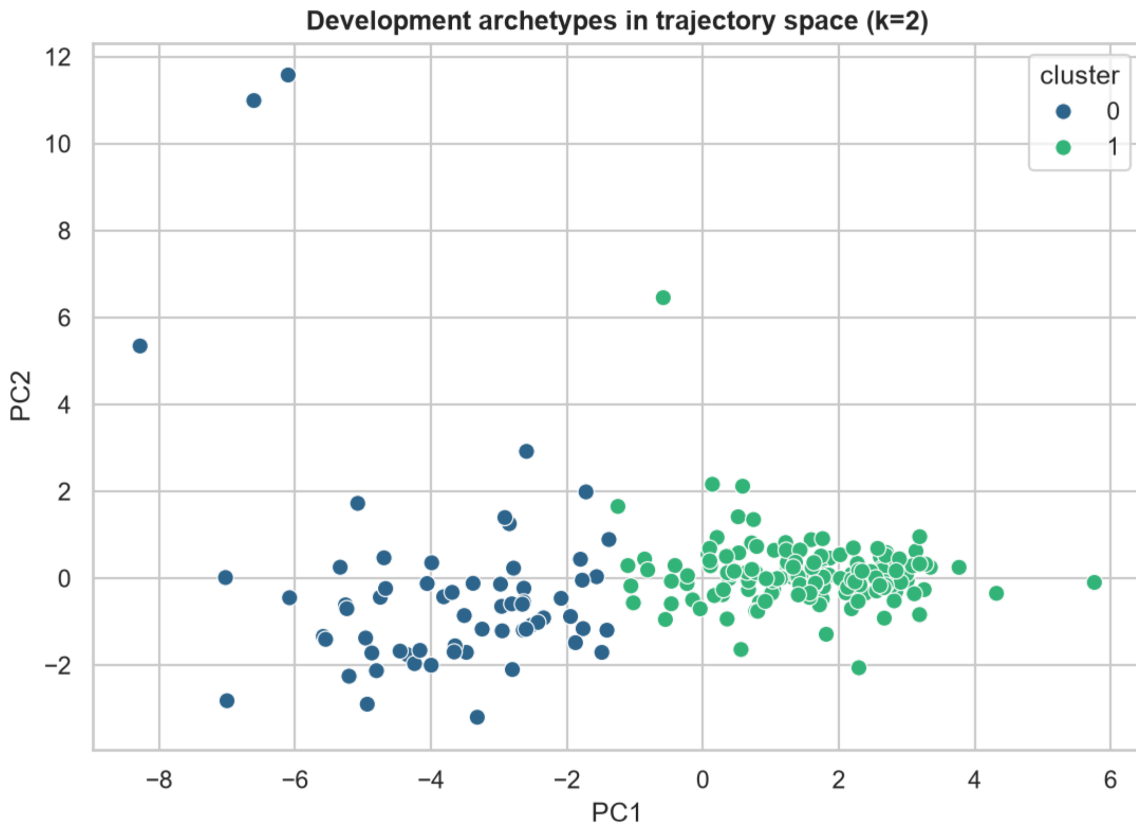


Figure 8. Choosing k. The silhouette peaks at **k = 2** (0.312), essentially tied with k = 3 (0.300) before falling sharply. We adopt **k = 2** as the primary, most *stable* solution (bootstrap ARI = 0.96 ± 0.05); K-means and Ward agree strongly (ARI = 0.82). Forcing k = 3 splits off a small **fragile-states** group (Central African Republic, Somalia, South Sudan) but with markedly lower stability (ARI ≈ 0.78), so we treat it as exploratory.



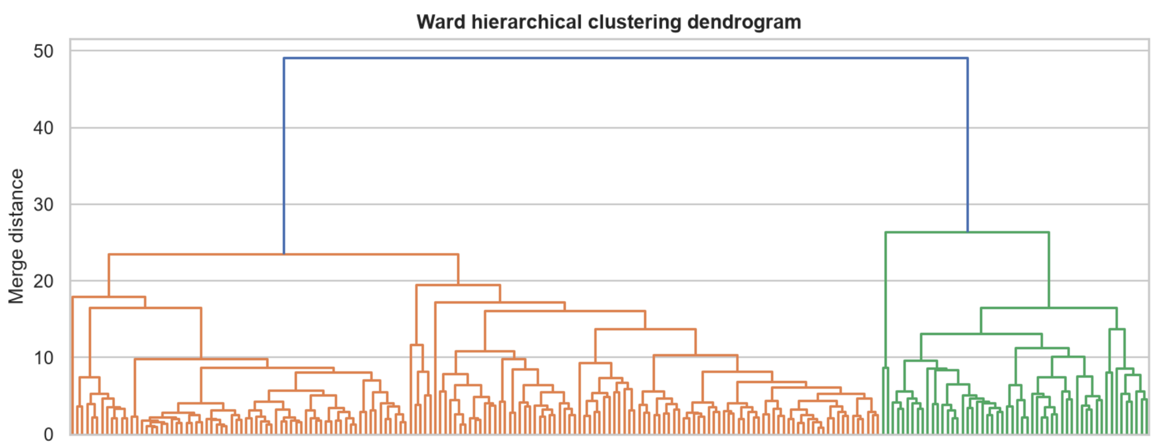


Figure 9. Archetypes in trajectory space. The two clusters separate cleanly along PC1 (the dominant "development level" axis). Cluster 0 is the **emerging/developing** archetype, cluster 1 the **advanced** archetype.

cluster	n_countries	life_expectancy__level	life_expectancy__slope	gdp_pc_ppp__level	under5_mortality__level
0	64	60.9	4.5	5173.8	75.6
1	153	75.8	1.8	32755.0	13.9

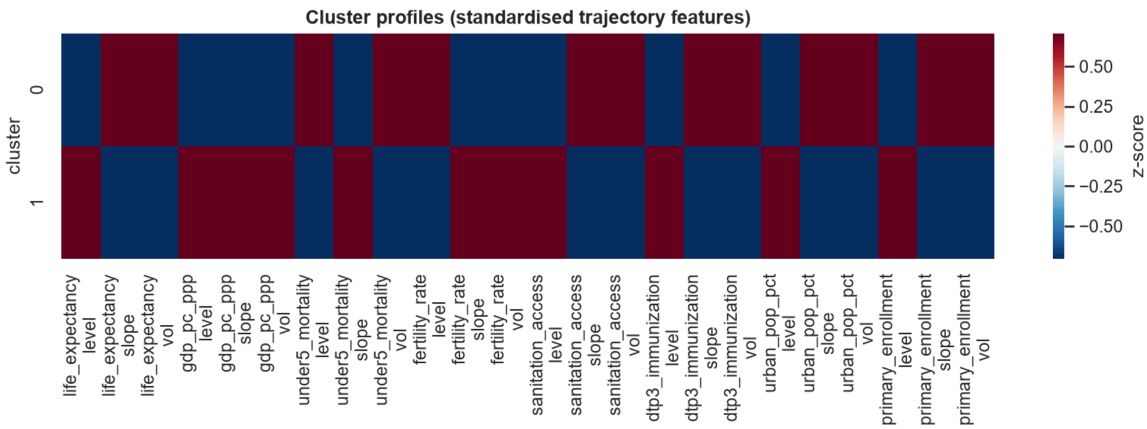
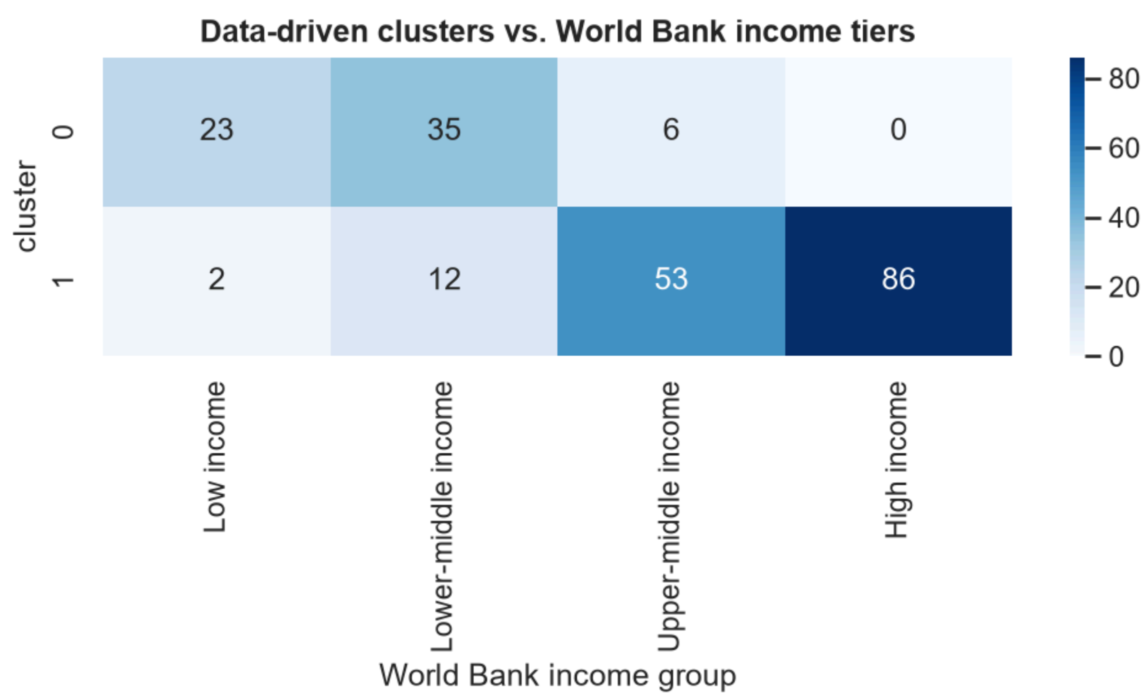


Figure 10 & Table 4. Cluster profiles. Beyond the obvious level gaps (advanced cluster: life expectancy ≈ 76 vs. 61 years; GDP ≈ $32.8kvs.$5.2k), the **slope** tells the interesting story: the emerging cluster is improving its life expectancy about **2.5× faster** (+4.5 vs. +1.8 years/decade) and expanding sanitation far faster — classic *convergence* that a static income tier cannot express.

3.6 Clusters vs. income tiers, and who diverges

Purpose. Cross-tabulate the data-driven clusters against World Bank income groups and surface the countries whose archetype disagrees with their income class.




```
Under-performers (rich income tier, low-development cluster):
```

	name	income_group	cluster	gap
0	Botswana	Upper-middle income	0	-0.666667
1	Micronesia, Fed. Sts.	Upper-middle income	0	-0.666667
2	Gabon	Upper-middle income	0	-0.666667
3	Equatorial Guinea	Upper-middle income	0	-0.666667
4	Iraq	Upper-middle income	0	-0.666667
5	South Africa	Upper-middle income	0	-0.666667
6	Angola	Lower-middle income	0	-0.333333
7	Benin	Lower-middle income	0	-0.333333
8	Bangladesh	Lower-middle income	0	-0.333333
9	Bolivia	Lower-middle income	0	-0.333333

```
Over-performers (low income tier, high-development cluster):
```

	name	income_group	cluster	gap
0	Korea, Dem. People's Rep.	Low income	1	1.000000
1	Syrian Arab Republic	Low income	1	1.000000
2	Egypt, Arab Rep.	Lower-middle income	1	0.666667
3	Honduras	Lower-middle income	1	0.666667
4	Kyrgyz Republic	Lower-middle income	1	0.666667
5	Lebanon	Lower-middle income	1	0.666667
6	Morocco	Lower-middle income	1	0.666667
7	Nicaragua	Lower-middle income	1	0.666667
8	West Bank and Gaza	Lower-middle income	1	0.666667
9	Tajikistan	Lower-middle income	1	0.666667

Figure 11 & Tables 5–6. Divergence from income tiers (H2). The clusters broadly track income but with revealing exceptions. **Under-performers** — upper-middle- income countries that cluster with the developing world — are dominated by **resource-rich, high-inequality economies** (Equatorial Guinea, Gabon, Botswana, South Africa, Iraq): high GDP that has not translated into commensurate health and social development, exactly the "GDP is not enough" cases. **Over-performers** — lower-income countries clustering with the advanced group — have relatively strong health/education systems for their income (e.g. Kyrgyz Republic, Morocco, Nicaragua, West Bank & Gaza), good candidates for policy case studies. (North Korea and Syria also appear here but reflect known data-quality limits.) This confirms **H2**: trajectory clusters are *not* a relabelling of income tiers.

4. Limitations

We flag the main threats to validity and scope so the results are read in context.

- **Data quality and coverage.** WDI values contain occasional source errors (some of which our despiking removes, but not all) and the sparsest series — the Gini index and the Gallup-sourced social variables — are missing for a large share of country-years, so parts of the feature space rely on imputation. A few countries (notably North Korea and Syria) have known reliability issues yet still surface in the cluster analysis.
- **Imputation caveat.** MICE is fit inside CV folds (no leakage), but the prior within-country interpolation uses a country's own past *and* future values; it is harmless under grouped CV (a country sits in one fold) yet mildly optimistic for the temporal hold-out.
- **Hyperparameter selection.** Grids are tuned with grouped CV and then the same data is used for the out-of-fold evaluation; a fully **nested** CV would remove the small optimistic bias this introduces. Search grids are modest by design.
- **Association, not causation.** Permutation and SHAP importances identify which indicators *precede* and *predict* outcomes; they are **not** causal effects. Confounding (e.g. governance quality driving many indicators at once) is not addressed.
- **Correlated features.** Health, economic and social indicators are highly inter-correlated, so importance is shared among proxies (e.g. under-5 vs. infant mortality) and individual rankings should be read as *groups* of related signals.
- **Horizon and panel length.** With a 4-year horizon over 2005–2020 each country contributes a limited number of ($t \rightarrow t+4$) windows; the negative gain-R^2 for happiness partly reflects this short, annual panel rather than pure unpredictability.
- **Clustering choices.** Trajectory features (level/slope/volatility) and *k* are design decisions; the silhouette-optimal $k = 2$ is robust, but the finer $k = 3$ structure is less stable and should be treated as exploratory.

5. Conclusions

On the problem. (1) Adding health and social information to economic inputs substantially improves forecasts of future well-being — roughly halving life-expectancy forecast error and lifting happiness R^2 by ~0.12 — confirming *H1*. Feature attribution localises the effect: **child-health and sanitation** drive future *longevity*, while **social support** (more than income) drives future *happiness*. (2) Clustering countries on trajectory shape yields a robust developed/developing split whose disagreements with income tiers are substantively meaningful: resource-rich under-performers and social-capital-rich over-performers, plus a *momentum* dimension (fast-converging developing countries) that income tiers miss entirely — confirming *H2*.

On machine learning in general.

- **Baselines discipline claims.** A trivial persistence forecast beats every learned model that ignores the lagged outcome; without it, an R^2 of 0.89 would look impressive rather than sobering.
- **Evaluation design changes the story.** Grouped-by-country CV (generalise to new countries) is far harder than a temporal hold-out (generalise to new years for known countries). Reporting a single split can be

misleading.

- **Negative results are informative.** Multi-year *gains* in happiness are essentially unpredictable from slow structural indicators — a finding, not a failure.
- **Interpretability matters more than a leaderboard.** XGBoost only modestly out-scores Ridge, but permutation/SHAP attribution is what actually answers the policy question.

Future work. (i) Extend the panel to 2023 and add lagged/rolling features and richer dynamics (e.g. time-series K-means with DTW on raw trajectories); (ii) model *gains* with explicit shock covariates (conflict, pandemics, commodity prices) to attack the negative gain-R^2; (iii) add uncertainty quantification (quantile or conformal prediction) for policy use; (iv) causal framing (e.g. panel fixed-effects or double machine learning) to move from "precedes" to "causes"; (v) cluster on the full WHR+WDI feature set for finer, well-being-aware archetypes.

6. Code and reproducibility

The full, commented implementation lives in `src/` and is orchestrated by `run_all.py`. See `README.md` for setup. In brief:

```
pip install -r requirements.txt
python run_all.py --rebuild   # re-download WDI + WHR, rebuild panel, run everything
python build_report.py        # regenerate this notebook
```

Module map

File	Responsibility
`src/config.py`	paths, indicator lists, feature groups, pipeline parameters
`src/data_acquisition.py`	pull WDI (`wbgapi`) + WHR, reconcile names, merge panel
`src/preprocessing.py`	despiking, within-country interpolation, t→t+k targets, feature sets
`src/prediction.py`	model zoo, grouped CV, hypothesis test, permutation/SHAP, temporal split
`src/clustering.py`	trajectory features, k-selection, K-means/Ward, stability, income cross-tab
`src/utils.py`	metrics, leakage-safe model pipelines, plotting
`run_all.py`	end-to-end pipeline → `figures/` + `results/`

All randomness is seeded (`RANDOM_STATE = 42`). Evaluation uses grouped-by-country `GroupKFold`; imputation and scaling are fit inside CV folds to prevent leakage.

```
H1 — life expectancy level R^2 :  econ-only 0.726  ->  full 0.922
H1 — life-ladder   level R^2 :  econ-only 0.581  ->  full 0.721
H2 — clusters k=2, silhouette=0.312, bootstrap ARI=0.958
```